

Integral-equation analysis of transient diffusion-limited currents at disk electrodes: Asymptotic expansion and compact approximation

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Abstract

The transient diffusion-limited current at a disk electrode following a change in interfacial ion concentration induced by a potential step is analyzed with direct relevance to chronoamperometric measurements. The mixed-boundary diffusion problem is formulated in the Laplace domain and reduced to a Fredholm integral equation that directly determines the Faradaic current. The steady-state limit recovers Saito's equation, while a systematic long-time asymptotic expansion quantifies the approach to steady state. A Padé approximant yields a compact analytical expression in the time domain that accurately describes the current over experimentally relevant time ranges. In contrast to existing high-accuracy numerical procedures based on hybrid asymptotic and polynomial approximations, the present formulation provides an explicit and compact analytical representation that facilitates interpretation and practical implementation. The short-time response exhibits Cottrell's equation with edge effects characteristic of disk electrodes. Overall, the framework provides practical tools for analyzing transient currents, extracting diffusion parameters, and assessing the accuracy of widely used analytical approximations in disk-electrode chronoamperometry.

I. INTRODUCTION

Disk electrodes, particularly in the ultramicroelectrode regime, are widely used in electrochemical measurements because they provide high current densities, rapid attainment of steady-state conditions, and reduced ohmic and capacitive effects. These characteristics make disk and microdisk electrodes especially suitable for quantitative investigations of mass transport, electrode kinetics, and coupled reaction–diffusion processes, for the detection of low-concentration species, and for a broad range of analytical and bioelectrochemical applications.^{1–17} These characteristics arise from enhanced radial diffusion toward the electrode, which increases the flux relative to planar geometries, and from the small electrode size, which minimizes ohmic drop and double-layer charging currents.

At the same time, ultramicroelectrodes present certain limitations. In particular, the absolute currents are small, which can impose stringent requirements on instrumentation and signal-to-noise ratios. In addition, experimental implementation may be more demanding, as the fabrication and characterization of well-defined microelectrode geometries require careful control. Variations in electrode surface condition and geometry can also affect reproducibility, particularly in quantitative measurements.

Following a potential step, the chronoamperometric current at a disk electrode relaxes from an initial transient regime toward a finite steady-state value. At very short times, the total current follows Cottrell’s equation, exhibiting $t^{-1/2}$ time dependence, where t denotes time, a behavior also observed for other electrode geometries, including spherical electrodes. At longer times, however, the disk geometry gives rise to a finite steady-state current that reflects the mixed boundary conditions imposed by a reactive circular region embedded in an otherwise insulating plane.

From a theoretical standpoint, diffusion to a disk electrode constitutes a mixed boundary-value problem, with Dirichlet boundary conditions applied on the disk surface and Neumann (insulating) boundary conditions imposed on the surrounding plane. Such mixed boundary conditions lead to nonuniform current-density distributions and edge-related singular behavior, particularly in the short- and intermediate-time regimes. Analytical studies of diffusion out-

side conducting bodies have shown that edge effects introduce systematic asymptotic corrections to simple one-dimensional diffusion models, even when the total current follows Cottrell’s equation.^{5,18–20}

For practical data analysis, several approximate analytical expressions have been proposed to describe the transient diffusion-limited current at disk electrodes.^{21–25} Among these, the Shoup–Szabo equation has become the most widely used interpolation formula, as it provides a convenient closed-form expression that smoothly connects the short- and long-time limits.²² Despite its practical utility, experimental and numerical studies have shown that the accuracy of this approximation depends on the time window selected for fitting, with the largest deviations typically occurring in the intermediate-time regime most commonly accessed in chronoamperometric experiments.⁸

Exact analytical solutions for diffusion to disk electrodes can be formulated in terms of special functions, such as spheroidal wave functions; however, their mathematical complexity and the need for numerical evaluation of infinite series limit their direct applicability in routine data analysis.^{25–27} Although a formal exact solution exists in the Laplace domain, its practical evaluation requires numerical treatment of special functions and infinite series. At the same time, highly accurate and computationally efficient numerical procedures have been developed. In particular, the approach reported in Ref. 27 enables the evaluation of the transient current with near machine precision at very low computational cost by combining asymptotic expansions with polynomial approximations based on reference data. While such methods are extremely effective for numerical evaluation, they are inherently algorithmic and do not yield compact closed-form expressions that can be readily used for analytical manipulation, parameter estimation, or theoretical interpretation. This situation motivates the development of theoretical frameworks that treat the mixed boundary conditions rigorously while remaining amenable to numerical evaluation and systematic approximation. In particular, there remains a need for simple analytical representations that provide transparent functional forms and can be readily incorporated into theoretical models and data analysis procedures. Such approaches are particularly valuable in modern micro- and nanoelectrode applications, where accurate interpretation of transient currents is essential for establishing mass-transport control.^{28–30} The

results obtained in the present work are consistent with high-accuracy numerical evaluations such as those reported in Ref. 27, which serve as a benchmark for assessing the present analytical approximations. The present approach is therefore complementary to high-accuracy numerical schemes, providing analytically tractable expressions with clear physical interpretation.

In the present work, we analyze the relaxation of the diffusion-limited current at a disk electrode of radius a following a change in interfacial ion concentration induced by a potential step at $t = 0$. By formulating the diffusion problem in the Laplace domain, the governing equations are reduced to a modified Helmholtz equation and subsequently to a Fredholm integral equation of the second kind for an auxiliary function that directly determines the total Faradaic current. This formulation enables accurate numerical evaluation of the transient current and the systematic derivation of long-time asymptotic expansions. In addition, a Padé resummation in the Laplace domain is introduced to obtain a compact closed-form approximation that accurately describes the current over intermediate and long times.

It is important to distinguish between the formally exact short-time limit and the transient regimes that are experimentally accessible in chronoamperometric measurements at disk electrodes. An idealized instantaneous potential step leads, in the limit $t \rightarrow 0^+$, to a Cottrell-type current proportional to the electrode area, reflecting locally planar diffusion. In practice, however, this regime is often obscured by finite instrumental response times, double-layer charging, and other non-diffusive effects.^{5,31} Consequently, the experimentally relevant transient response is typically governed by the approach to the finite steady-state current of a disk electrode, where the leading correction again exhibits a Cottrell-type decay with $t^{-1/2}$ time dependence and a distinct amplitude. This regime underlies widely used analytical approximations for disk electrodes and motivates a unified theoretical description that consistently connects the short-time, long-time, and steady-state limits.

Overall, the approach presented here provides a unified and mathematically transparent description of steady-state, transient, and short-time behavior at disk electrodes, clarifies the role of edge effects arising from mixed boundary conditions, and offers practical analytical and numerical tools for interpreting chronoamperometric data beyond existing interpolation formulas.

II. FORMULATION

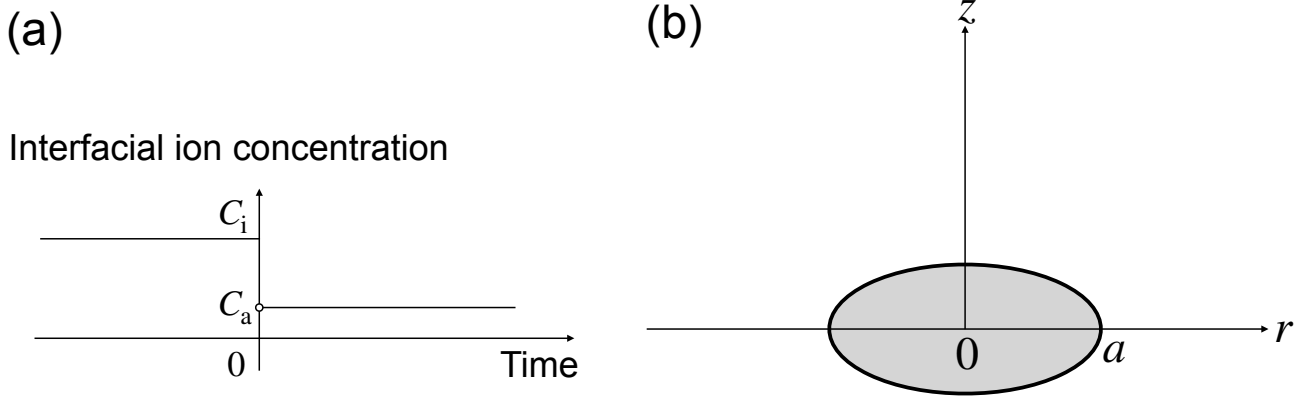


FIG. 1. (a) Initial uniform concentration C_i . At $t = 0^+$, the interfacial ion concentration is changed to C_a by applying a potential step and is maintained at this value thereafter. We define $\Delta C(r, z, t) = C(r, z, t) - C_i$ and $C_0 = C_a - C_i$. (b) Cylindrical coordinate system (r, z) used for diffusion modeling and flux/current calculations at a disk electrode of radius a . If $I(t) < 0$, the current flows in the negative z -direction.

We consider a system that is initially under non-reactive conditions for $t < 0$, with a uniform concentration C_i (Fig. 1(a)). At $t = 0^+$, the interfacial ion concentration is changed to C_a by applying a potential step. We study the subsequent current relaxation when the concentration $C(r, 0, t)$ within the circular region of radius a at $z = 0$ is maintained at C_a for $t > 0$.

Aside from the initial charging of the electric double layer, the current $I(t)$ is diffusion-controlled. We impose a Dirichlet boundary condition at the electrode surface, $C(r, 0, t) = C_a$, which corresponds to sufficiently fast electron-transfer kinetics (Nernstian equilibrium) at the interface. Under this condition, the surface concentration is fixed by thermodynamic equilibrium when $C_a > 0$, and transport remains diffusion-limited. In addition, no homogeneous chemical reaction between reactant and product species in the solution is assumed, except in Sec. 9, where the relation to a diffusional EC' mechanism is discussed. Accordingly, a single diffusion equation provides an adequate description. In the limiting case of a fast irreversible reaction, $C_a \approx 0$. The diffusion-controlled current $I(t)$ relaxes to the steady-state value I_∞ .

We examine the relaxation function $I(t)/I_\infty$, which is independent of C_i and C_a owing to the linearity of the diffusion equation. For convenience, we define $\Delta C(r, z, t) = C(r, z, t) - C_i$.

We analyze the transport of charge carriers to a circular reactive region (disk) of radius a located in the plane $z = 0$. The plane $z = 0$ is insulating except for the disk; that is, there is no normal flux across $z = 0$ outside the reactive area. The origin is placed at the disk center, and r denotes the radial coordinate in the plane $z = 0$ (Fig. 1(b)). The concentration field $C(r, z, t)$ is considered in the upper half-space, $z \geq 0$. Assuming rotational symmetry about the z -axis, $C(r, z, t)$ satisfies Fick's second law,

$$\frac{\partial C(r, z, t)}{\partial t} = D \nabla^2 C(r, z, t), \quad (1)$$

where the Laplacian under rotational symmetry is given by

$$\nabla^2 C(r, z, t) = \Delta_2 C(r, z, t) + \frac{\partial^2 C(r, z, t)}{\partial z^2}, \quad (2)$$

with

$$\Delta_2 C(r, z, t) = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C(r, z, t)}{\partial r} \right). \quad (3)$$

We denote the Laplace transform of $f(t)$ by $\hat{f}(s)$. As shown in Appendix A, the concentration in the upper half-space under the circular boundary conditions at $z = 0$ can be expressed as³²

$$\Delta \hat{C}(r, z, s) = \int_0^\infty dk A(k, s) J_0(kr) \exp\left(-z \sqrt{k^2 + s/D}\right), \quad (4)$$

where k is the separation constant arising from the method of separation of variables and serves as the radial transform variable in the integral representation of the solution, $A(k, s)$ is determined by the boundary conditions, and $J_n(z)$ denotes the Bessel function of the first kind³³. By definition, $\Delta C(r, z, t) \rightarrow 0$ as $r \rightarrow \infty$ and/or $z \rightarrow \infty$; thus, $C(r, z, t) \rightarrow C_i$ in these limits, where the influence of the circular electrode becomes negligible.

Faradaic current

To avoid confusion between diffusive flux and electrical current, we use $\vec{J}$ for the diffusive molar flux density (units of amount area⁻¹ time⁻¹) and I for the total Faradaic current (units

of charge time^{-1}). The Faradaic conversion is performed by the factor nF , where F indicates the Faraday constant, n is the number of electrons transferred.

We define the diffusive flux density vector,

$$\hat{\vec{J}} = -D \text{grad} \Delta \hat{C}(r, z, s). \quad (5)$$

The flux in the z -direction can be expressed as

$$\hat{J}_z(r, z, s) = -D \frac{\partial}{\partial z} \Delta \hat{C}(r, z, s). \quad (6)$$

By substituting Eq. (4) into Eq. (6), we obtain

$$\lim_{z \rightarrow +0} \hat{J}_z(r, z, s) = -D \frac{\partial}{\partial z} \int_0^\infty dk A(k, s) J_0(kr) \exp\left(-z\sqrt{k^2 + s/D}\right) \Big|_{z=+0} \quad (7)$$

$$= D \int_0^\infty dk \sqrt{k^2 + s/D} A(k, s) J_0(kr). \quad (8)$$

The total Faradaic current through the circular region at $z = 0$ can be calculated from

$$\hat{I}(s) = 2\pi nF \int_0^a r dr \hat{J}_z(r, z = +0, s) \quad (9)$$

$$= 2\pi D nF \int_0^a r dr \int_0^\infty dk \sqrt{k^2 + s/D} A(k, s) J_0(kr) \quad (10)$$

$$= 2\pi D a nF \int_0^\infty dk \frac{\sqrt{k^2 + s/D}}{k} A(k, s) J_1(ka), \quad (11)$$

where we have used (10.22.1 of Ref. 33)

$$\int_0^a dr r J_0(kr) = \frac{a}{k} J_1(ka). \quad (12)$$

If $I(t)$ is negative, the current flows in the negative direction of the z -axis in the upper half-space, $z \geq 0$.

Introducing $C_0 = C_a - C_i$ and $\Delta C(r, z, t) = C(r, z, t) - C_i$, the boundary condition inside the disk is expressed as $\Delta C(r, 0, t) = C_0$ for $0 \leq r < a$. Outside the disk, the plane is insulating; that is, $\partial \Delta C(r, z, t) / \partial z = 0$ at $z = 0$ for $r > a$. In the Laplace domain, $\Delta \hat{C}(r, z, s)$ satisfies the boundary conditions

$$\Delta \hat{C}(r, 0, s) = \frac{C_0}{s} \quad \text{for } 0 \leq r < a, \quad (13)$$

$$\lim_{z \rightarrow +0} \hat{J}_z(r, z, s) = 0 \quad \text{for } a < r < \infty. \quad (14)$$

Using Eqs. (4) and (8), we note that $A(k, s)$ satisfies the dual integral equations implied by these boundary conditions,³²

$$\int_0^\infty dk A(k, s) J_0(kr) = C_0/s \text{ for } 0 \leq r < a, \quad (15)$$

$$\int_0^\infty dk \sqrt{k^2 + s/D} A(k, s) J_0(kr) = 0 \text{ for } a < r < \infty. \quad (16)$$

To reduce these equations to a form for which a standard solution method is available, we introduce $\hat{g}(k, s)$ satisfying³²

$$\hat{g}(k, s) = \sqrt{k^2 + s/D} A(k, s)/C_0, \quad (17)$$

and rewrite Eqs. (15) and (16) as³²

$$\int_0^\infty dk \frac{\hat{g}(k, s)}{\sqrt{k^2 + s/D}} J_0(kr) = 1/s \text{ for } 0 \leq r < a, \quad (18)$$

$$\int_0^\infty dk \hat{g}(k, s) J_0(kr) = 0 \text{ for } a < r < \infty. \quad (19)$$

A method for solving the dual integral equations [Eqs. (18) and (19)] has been established^{32,34–37}. Although Eqs. (18) and (19) can be interpreted as inverse Hankel transforms, a more effective method was proposed by Cooke. According to Cooke (1956) (see p. 213 of Ref. 32)³⁴, we express $\hat{g}(k, s)$ in terms of $\hat{h}(r, s)$ as

$$\hat{g}(k, s) = \frac{2}{\pi} k \int_0^a \hat{h}(r, s) \cos(kr) dr. \quad (20)$$

We show below that $\hat{h}(r, t)$ gives the z -component of the flux at the boundary, and hence determines the Faradaic current.

The total current flowing through the circular region of radius a at $z = 0$, Eq. (11), can be expressed as

$$\hat{I}(s) = 2\pi D a n F C_0 \int_0^\infty dk \hat{g}(k, s) \frac{1}{k} J_1(ka). \quad (21)$$

Using $\hat{h}(r, s)$, $\hat{I}(s)$ can be expressed as

$$\hat{I}(s) = 4D a n F C_0 \int_0^a dr \hat{h}(r, s) \int_0^\infty dk J_1(ka) \cos(kr) \quad (22)$$

$$= 4D n F C_0 \int_0^a dr \hat{h}(r, s), \quad (23)$$

where we have used $\int_0^\infty dk J_1(ka) \cos(kr) = 1/a$ for $0 < r < a$. (6.693.1 of Ref. 38) Equation (23) indicates that $h(r, t)$ determines the radial distribution of the flux contribution to the total Faradaic current at position r on the boundary $z = 0$.

III. STEADY STATE

Results in the steady state ($t \rightarrow \infty$) can be obtained by taking the limit $s \rightarrow 0$ and applying the inverse Laplace transformation; we note that the inverse Laplace transform of $1/s$ is 1.

In the limit $s \rightarrow 0$, we define

$$\hat{g}_0(k) = \lim_{s \rightarrow 0} \sqrt{k^2 + s/D} A(k, s)/C_0 = kA(k, 0)/C_0. \quad (24)$$

By taking the limit $s \rightarrow 0$, and applying the inverse Laplace transformation, Eqs. (18) and (19) become

$$\int_0^\infty dk \frac{g_0(k)}{k} J_0(kr) = 1 \text{ for } 0 \leq r < a, \quad (25)$$

$$\int_0^\infty dk g_0(k) J_0(kr) = 0 \text{ for } a < r < \infty. \quad (26)$$

Using the formula (6.693.6 and 6.671.7 of Ref. 38)

$$\int_0^\infty \frac{dk}{k} \sin(ka) J_0(kr) = \pi/2 \quad \text{for } 0 \leq r < a, \quad (27)$$

$$\int_0^\infty dk \sin(ka) J_0(kr) = 0 \quad \text{for } a < r < \infty, \quad (28)$$

we obtain

$$g_0(k) = \frac{2}{\pi} \sin(ka). \quad (29)$$

The analytical expression for the total current flowing through the circular region of radius a [the inverse Laplace transform of Eq. (21)] is then

$$I_\infty = 2\pi D a n F C_0 \int_0^\infty dk g_0(k) \frac{1}{k} J_1(ka) = 4 D a n F C_0, \quad (30)$$

where we have used $\int_0^\infty dk \sin(ka) J_1(ka)/k = 1$. (6.693.1 of Ref. 38) Equation (30) corresponds to Saito's equation.³⁹ When formulated in terms of electrical or ionic conductivity (or, equivalently, resistivity), this expression is recognized as the Holm constriction (spreading) resistance

or, alternatively, the Maxwell–Hall access resistance.^{40–44} (For the rate of heat flow, see Sec. 8.2 of Ref. 45) This expression has also been extended to describe anisotropic layers in the context of electrical conductivity.^{46–51} Here, we apply the method of solving mixed boundary value problems, originally developed to analyze electrical contacts in anisotropic layers, to the relaxation of diffusive currents toward the steady-state current described by Saito’s equation.

IV. TIME DEPENDENT SOLUTION: DERIVATION

We study the transient relaxation of the current to the steady state given by Eq. (30), *i.e.*, Saito’s equation. The results of this section are a special case of the general solution given in the Appendix C. Nevertheless, we provide the derivation for completeness. We also reformulate the problem using dimensionless variables in Appendix C.

We first substitute Eq. (20) into Eq. (19) by rewriting Eq. (20) as $\hat{g}(k, s) = (2/\pi)k \int_0^a dy \hat{h}(y, s) \cos(ky)$,

$$\int_0^a dy \hat{h}(y, s) \int_0^\infty dk \frac{2}{\pi} k \cos(ky) J_0(kr) = 0 \text{ for } a < r < \infty. \quad (31)$$

The boundary condition given by Eq. (31) is satisfied irrespective of the functional form of $\hat{h}(r, s)$, since $\hat{h}(r, s)$ is independent of k and

$$\int_0^\infty dk k \cos(ky) J_0(kr) = 0 \text{ for } y < r < \infty, \quad (32)$$

which follows from (10.22.59 of Ref. 33)

$$\int_0^\infty dk \sin(ky) J_0(kr) = 0 \text{ for } y < r < \infty; \quad (33)$$

we then have

$$\frac{d}{dy} \int_0^\infty dk \sin(ky) J_0(kr) = 0 \text{ for } y < r < \infty. \quad (34)$$

Noting that $y < r$ in Eq. (31), we obtain Eq. (32). We have thus shown that the boundary condition in Eq. (19) is satisfied irrespective of the functional form of $\hat{h}(r, s)$.

Next, we substitute Eq. (20) into Eq. (18),

$$\frac{2}{\pi} \int_0^a dy \hat{h}(y, s) \int_0^\infty dk \cos(ky) \frac{k}{\sqrt{k^2 + s/D}} J_0(kr) = 1/s \text{ for } 0 \leq r < a, \quad (35)$$

The function $\hat{h}(r, s)$ satisfying the other boundary condition given by Eq. (18) must satisfy the Fredholm integral equation of the first kind for $\hat{h}(y, s)$, given by Eq. (35).

As shown in Appendix B, Eq. (35) can be rewritten as a Fredholm integral equation of the second kind for $\hat{h}(y, s)$ ³²

$$\hat{h}(r, s) = \frac{1}{s} + \int_0^a dy \hat{h}(y, s) \hat{K}(r, y, s) \quad \text{for } 0 \leq r < a, \quad (36)$$

where the kernel is defined as

$$\hat{K}(r, y, s) = -\frac{2}{\pi} \int_0^\infty dk \left[\frac{k}{\sqrt{k^2 + s/D}} - 1 \right] \cos(ky) \cos(kr) \quad \text{for } 0 \leq r < a. \quad (37)$$

Equations (20) and (36), together with Eq. (37), can be derived by applying the general method described in Appendix C, *i.e.*, Eqs. (C.7) and (C.8).

Since the total current is obtained by spatial integration of $h(r, t)$ over the disk area, $I(t) = 4DnFC_0 \int_0^a dr h(r, t)$ [Eq. (23)], $h(r, t)$ is proportional to the local contribution to the flux-weighted current density normal to the disk surface at radial position r and time t . The first term on the right-hand side of Eq. (36) corresponds to the steady-state current, while the integral term describes the temporal evolution toward this steady state. Specifically, the integral term can be interpreted as a memory effect; the local flux at position r and time t depends on the distribution of fluxes at earlier times $t' < t$ over the entire disk.

This nonlocal temporal coupling is mediated by the kernel $K(r, y, t - t')$, which quantifies how a current density at position y and time t' contributes to the response at position r at a later time t . In the Laplace domain, this memory kernel is represented by $\hat{K}(r, y, s)$. Thus, Eq. (36) explicitly expresses the relaxation process as a spatio-temporal convolution, reflecting the diffusive coupling between different regions of the electrode surface.

By substituting $\hat{h}(r, s)$ determined by solving Eq. (36) with Eq. (37) into Eq. (23), we obtain the total current flowing through the circular region of radius a at $z = 0$ in the Laplace domain. To calculate $\hat{I}(s)$, we need $\hat{h}(r, s)$ in the integrand of $\int_0^a dr \dots$, which indicates that $\hat{h}(r, s)$ for $0 \leq r < a$ is required. Therefore, we drop the condition $0 \leq r < a$ for $\hat{h}(r, s)$ below. Equation (36) involves double integrations. However, Eq. (37) can be expressed by analytical

function,⁵²

$$\hat{K}(r, y, s) = \frac{1}{2} \sqrt{\frac{s}{D}} \left[\frac{4}{\pi} + \left(\mathbf{L}_1 \left(\sqrt{\frac{s}{D}}(r+y) \right) - I_1 \left(\sqrt{\frac{s}{D}}(r+y) \right) \right) + \left(\mathbf{L}_1 \left(\sqrt{\frac{s}{D}}(r-y) \right) - I_1 \left(\sqrt{\frac{s}{D}}|r-y| \right) \right) \right], \quad (38)$$

where $\mathbf{L}_n(z)$ indicates the modified Struve function, and $I_n(z)$ indicates the modified Bessel function of the first kind.³³ We note that $\mathbf{L}_1(z)$ is an even function.³³

Equations (20) and (36), together with Eq. (37), have already been reported in Ref. 32. By using Eq. (38), Eq. (36) can be discretized more straightforwardly than performing the numerical integration required in Eq. (37). In the following, we explicitly solve Eqs. (20) and (36) using Eq. (38). The numerical solution of Eq. (36) is substituted into Eq. (23); we obtain the Laplace transform of the total current flowing through the circular region of radius a at $z = 0$. The time dependence is obtained by applying a numerical inverse Laplace transform to Eq. (23) using the Stehfest algorithm.^{53,54}

In the following, we distinguish between the steady-state solution and the time-dependent asymptotic solution. From a formal mathematical perspective, the steady-state solution corresponds to the leading-order term in the large-time asymptotic expansion. However, we use the term “steady-state solution” to refer specifically to the strictly time-independent limit, whereas the “asymptotic solution” describes the time-dependent behavior approaching this limit.

V. STEADY STATE: REVISITED

We define $\hat{h}_0(r) = \lim_{s \rightarrow 0} \hat{h}(r, s)$, and $\hat{g}_0(r) = \lim_{s \rightarrow 0} \hat{g}(r, s)$. Using Eq. (36), we find $\hat{h}_0(r) = 1/s$ and $h_0(r) = 1$ after applying the inverse Laplace transformation. Substituting $\hat{h}_0(r) = 1/s$ into Eqs. (20) and (23), we obtain $\hat{g}_0(r) = (2/\pi) \sin(ka)/s$, $g_0(k) = (2/\pi) \sin(ka)$ [Eq. (29)], and $I_\infty = 4DanFC_0$. Therefore, Saito’s equation [Eq. (30)] is derived.

VI. LONG-TIME ASYMPTOTIC SOLUTION

We consider the exact integral equation, Eq. (36), where $\hat{K}(r, y, s)$ is the kernel defined in Eq. (37). We first expand $\hat{K}(r, s)$ using Eq. (38) by regarding sa^2/D as a small parameter and assuming $r/a \leq 1$ and $y/a \leq 1$; the kernel $\hat{K}(r, y, s)$ admits the series representation

$$\hat{K}(r, y, s) = \sum_{n=1}^{\infty} K^{(n)}(r, y) \left(\frac{s}{D} \right)^{n/2}, \quad (39)$$

with coefficient functions $K^{(n)}(r, y)$ given by

$$K^{(n)}(r, y) = \left[\frac{1 - (-1)^n}{2} a_{(n-3)/2} - \frac{1 + (-1)^n}{2} b_{n/2} \right] \times \left[(r+y)^{n-1} + \frac{1 - (-1)^n}{2} (r-y)^{n-1} + \frac{1 + (-1)^n}{2} |r-y|^{n-1} \right], \quad (40)$$

where we defined,

$$a_n = \frac{2^{2n+1} n! (n+1)!}{\pi (2n+1)! (2n+3)!}, \quad (41)$$

$$b_n = \frac{n}{2^{2n} (n!)^2}, \quad (42)$$

and $a_{-1} = 1/\pi$. The factors $[1 - (-1)^n]/2$ and $[1 + (-1)^n]/2$ select the even- and odd- n contributions, respectively.

Corresponding to the expansion in powers of $(s/D)^{1/2}$ in Eq. (39), the solution $\hat{h}(r, s)$ may admit the following asymptotic expansion,

$$\hat{h}(r, s) = \sum_{\ell=0}^{\infty} h^{(\ell)}(r) \frac{1}{s} \left(\frac{sa^2}{D} \right)^{\ell/2}. \quad (43)$$

The ℓ th-order iterated function $h^{(\ell)}(r)$ is obtained by substituting Eqs. (39) and (43) into the integral equation, yielding

$$h^{(\ell)}(r) = \sum_{n=1}^{\ell} \int_0^a dy h^{(\ell-n)}(y) K^{(n)}(r, y) / a^n, \quad \ell \geq 1, \quad (44)$$

with the initial condition $h^{(0)}(r) = 1$ and $K^{(n)}(r, y)$ given by Eq. (40). In Appendix D, this recursive relation is implemented efficiently using pre-stored moment integrals.

Using the iterated functions obtained from Eq. (44), $\hat{h}(r, s)$ is derived from Eq. (43). Applying the inverse Laplace transform via the Tauberian theorem,⁵⁵ while neglecting the even-power terms in sa^2/D that do not contribute to the asymptotic expansion in the time domain, yields

$$h(r, t) = 1 + \sum_{m=0}^{\infty} \frac{h^{(1+2m)}(r)}{\Gamma(1/2 - m)} \left(\frac{a^2}{Dt} \right)^{1/2+m}. \quad (45)$$

The transient current follows from Eq. (23) as

$$\begin{aligned} I(t) &= 4DnFC_0 \int_0^a dr h(r, t) \\ &= 4DanFC_0 \left[1 + \sum_{m=0}^{\infty} \frac{\int_0^a dr h^{(1+2m)}(r)}{a\Gamma(1/2 - m)} \left(\frac{a^2}{Dt} \right)^{1/2+m} \right], \end{aligned} \quad (46)$$

while the corresponding expression in the Laplace domain is

$$\hat{I}(s) = \frac{4DanFC_0}{s} \left[1 + \sum_{m=0}^{\infty} \frac{1}{a} \int_0^a dr h^{(1+2m)}(r) \left(\frac{sa^2}{D} \right)^{1/2+m} \right]. \quad (47)$$

For convenience, we introduce,

$$\frac{I^{(1+2\ell)}}{I_{\infty}} = \frac{1}{a} \int_0^a dr h^{(1+2m)}(r), \quad (48)$$

where $I_{\infty} = 4DanFC_0$.

As shown in Appendix D, Eqs. (46) and (47) reproduce the asymptotic results obtained by evaluating the exact expression for the current expanded in terms of radial spheroidal wave functions of the third kind,²⁷ and provide a unified framework for generating higher-order correction terms. In Appendix D, higher-order correction terms obtained using this approach are also presented.

By applying the systematic asymptotic expansion up to the 9th iteration from the steady state in the long-time limit, we obtain

$$\begin{aligned} \frac{I_{\text{ip}}(t)}{I_{\infty}} &= 1 + \frac{2a}{\pi^{3/2}(Dt)^{1/2}} + \left(\frac{1}{9} - \frac{1}{\pi^2} \right) \frac{4a^3}{(\pi Dt)^{3/2}} + \left(\frac{71}{450} - \frac{4}{\pi^2} + \frac{24}{\pi^4} \right) \frac{a^5}{\pi^{3/2}(Dt)^{5/2}} + \\ &\quad \left(\frac{71}{2205} - \frac{242}{135\pi^2} + \frac{80}{3\pi^4} - \frac{120}{\pi^6} \right) \frac{2a^7}{\pi^{3/2}(Dt)^{7/2}} + \\ &\quad \left(\frac{9523}{340200} - \frac{5396}{1701\pi^2} + \frac{266}{3\pi^4} - \frac{2800}{3\pi^6} + \frac{3360}{\pi^8} \right) \frac{a^9}{\pi^{3/2}(Dt)^{9/2}}. \end{aligned} \quad (49)$$

Here $I_\infty = 4DanFC_0$ [Eq. (30)] and $I_{ip}(t)$ indicates the results of inverse power series expansion. The first three terms on the right-hand side of Eq. (49) coincide with the known results derived by a different method;²² $1 + 2a/(\pi^{3/2}(Dt)^{1/2})$ was derived by Shoup and Szabo by correcting an error in the original derivation by Aoki and Osteryoung using the Wiener–Hopf method,^{21–23} and the third term $(1/9 - 1/\pi^2) 4a^3/(\pi Dt)^{3/2}$ was also derived by Shoup and Szabo.²² The 4th term has been shown previously by applying a scattering method. (Eq. (29) of ref. 25)

VII. RESULTS

In the numerical calculations, Eq. (36) is discretized, together with Eq. (38), using a standard scheme to obtain $\hat{h}(r, s)$. The resulting $\hat{h}(r, s)$ is then substituted into Eq. (23) to evaluate the Laplace-domain total current through the circular region of radius a at $z = 0$. The time-dependent current is obtained by numerically inverting $\hat{I}(s)$ using the Stehfest algorithm.^{53,54}

Convergent values are obtained from Eqs. (44) and (46) by retaining terms up to $t^{-15/2}$ at $Dt/a^2 = 1$. For $Dt/a^2 > 1$, fewer recursive terms are required to reach convergence. We note, however, that including terms beyond $t^{-15/2}$ tends to increase the predicted current; thus, the convergence is effectively asymptotic. Nevertheless, the resulting values agree closely with the numerical calculations, as shown in Table I. The numerical results reported here are consistent with high-accuracy evaluations available in the literature; for example, the numerical coefficients in Eq. (D.8) are in exact agreement with those reported in Table 1 of Ref. 27. These results therefore serve as a benchmark for assessing the present analytical approximations. The results are summarized in Table I. Both the analytical and numerical values show close agreement with Eq. (6) of Mahon and Oldham.²⁴

In Fig. 2, we examine whether inverse power expansions converge in the Laplace domain and in the time domain. As shown in the figure, $I^{(1+2\ell)}/I_\infty$ in Eq. (48), which is the coefficient of $(sa^2/D)^{1/2+\ell}$, converges with increasing ℓ . In contrast, $I^{(1+2\ell)}/I_\infty/\Gamma(1/2 - \ell)$, which is the coefficient of $(Dt/a^2)^{-1/2-\ell}$, decreases up to $\ell = 12$ but then increases for larger ℓ . This difference originates from the factor $1/\Gamma(1/2 - \ell)$ in the inverse power-series expansion in the time domain. In Table I, we present convergent values calculated up to $\ell = 7$; these results are

TABLE I. Time dependence of the normalized current $I(t)/I_0$ for $Dt/a^2 > 1$

Dt/a^2 ^a	High-accuracy numerical ^{bcd}	Eq. (52)	Eq. (6) of Mahon–Oldham ²⁴	Eq. (8) of Shoup–Szabo ²²
1	1.36586(1.3659)	1.36521	1.36625	1.374
1.69	1.27941(1.279)	1.2792	1.27941	1.285
2	1.25641	1.25627	1.25641	1.26148
3	1.20871	1.20865	1.20871	1.21245
5	1.16125	1.16124	1.16125	1.16373
10	1.1138	1.1138	1.1138	1.11516

^a Dimensionless time of Shoup–Szabo is $4Dt/a^2$.²²

^b Numerical results calculated using Eq. (36) together with Eq. (38), providing benchmark-level accuracy for the present comparisons.

^c Convergent results of Eqs. (44) and (46). For $Dt/a^2 = 1$, the recursively determined current up to the $t^{-15/2}$ time dependence is used. These results are consistent with high-accuracy numerical evaluations reported in the literature (e.g., Ref. 27).

^d The numerical values in parentheses indicate the results of the numerical simulations reported in Ref. 56 for $Dt/a^2 = 1$,^{44,56,57} and in Ref. 22 for $Dt/a^2 = 1.69$.

consistent with the numerical results obtained by discretization of Eq. (36).

As shown in Fig. 3, when the plots are extended to the time region $Dt/a^2 < 1$, $I(t)/I_\infty$ obtained using the inverse power expansion in time does not systematically approach the numerical results with increasing ℓ . The best agreement in the time domain for $Dt/a^2 \gtrsim 0.2$ is obtained for $I_{\text{ip}}(t)/I_\infty$ given by Eq. (49), which includes terms up to the $1/t^{9/2}$ time dependence.

Motivated by the convergence properties of the series expansion in the Laplace domain, we introduce a resummation method based on the Padé approximant to obtain a convergent analytical expression in the time domain. First, we expand the current in the Laplace domain using Eqs. (40)–(44),

$$\frac{\hat{I}(s)}{I_\infty} \approx \frac{1}{s} + \frac{2a}{\pi(Ds)^{1/2}} + \left(\frac{4}{\pi^2} - \frac{1}{3}\right) \frac{a^2}{D} - \left(\frac{1}{9} - \frac{1}{\pi^2}\right) \frac{8a^3 s^{1/2}}{\pi D^{3/2}}. \quad (50)$$

Equation (50) is obtained by performing a systematic expansion in terms of $\sqrt{s/D}a$, including

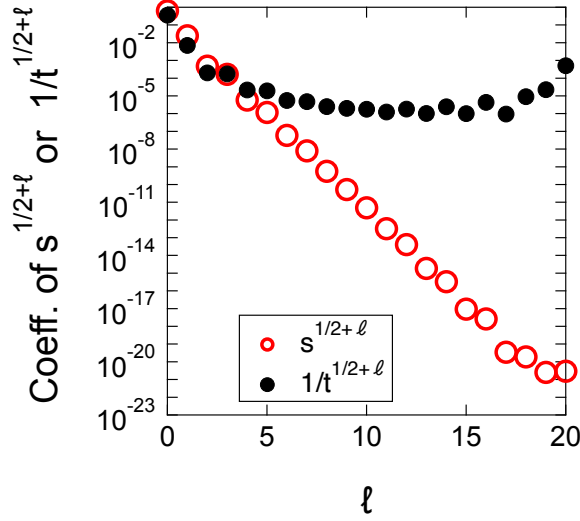


FIG. 2. Red open circles indicate the coefficients of $s^{1/2+\ell}$ in the expansion of the normalized current $I^{(1+2\ell)}/I_\infty$ as a function of ℓ in the Laplace domain. Black closed circles indicate the coefficients of $1/t^{1/2+\ell}$ in the expansion of the normalized current $I^{(1+2\ell)}/I_\infty/\Gamma(1/2-\ell)$ in the time domain.

a constant term in the Laplace domain, given by the third term on the right-hand side. Upon inverse Laplace transformation ($1 \leftrightarrow \delta(t)$), this constant term can be neglected when considering the asymptotic time dependence. Here, however, it is retained to interpolate the asymptotic behavior and extend the validity to short-time regimes. We then apply a Padé approximant of order [2,3],

$$\frac{\hat{I}(s)}{I_\infty} \approx \frac{3\pi(12-\pi^2) + 2\pi^2 a \sqrt{s/D} + \pi(\pi^2-8)a^2 s/D}{s \left[3\pi(12-\pi^2) + 8(\pi^2-9)a \sqrt{s/D} \right]}. \quad (51)$$

Taking the inverse Laplace transform yields the approximate analytical expression in the time domain,

$$\frac{I(t)}{I_\infty} \approx 1 + \frac{\sqrt{\pi}(\pi^2-8)}{8(\pi^2-9)} \frac{a}{\sqrt{Dt}} - \frac{3(12-\pi^2)^3}{64(\pi^2-9)^2} \exp\left(c_{\text{pd}}^2 \frac{Dt}{a^2}\right) \text{erfc}\left(c_{\text{pd}} \frac{\sqrt{Dt}}{a}\right), \quad (52)$$

$$\approx 1 + 0.476335 \frac{a}{\sqrt{Dt}} - 0.599347 \exp\left(c_{\text{pd}}^2 \frac{Dt}{a^2}\right) \text{erfc}\left(c_{\text{pd}} \frac{\sqrt{Dt}}{a}\right), \quad (53)$$

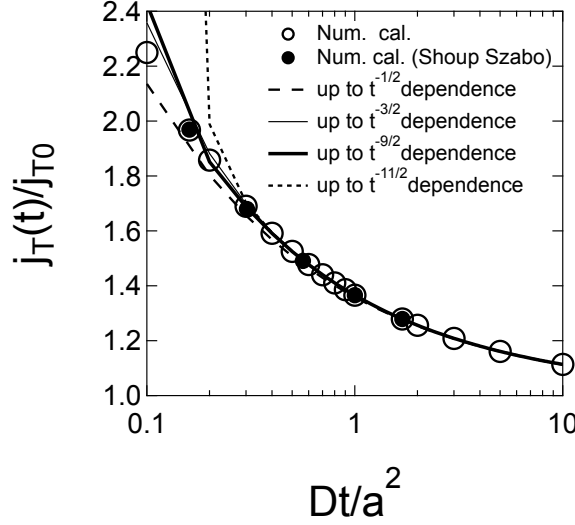


FIG. 3. Thin long dashed, thin solid, thick solid, and thin short dashed lines indicate $I_{ip}(t)/I_\infty$ obtained using inverse power-law expansions up to $1/t^{1/2}$, $1/t^{3/2}$, $1/t^{9/2}$ [Eq. (49)], and $1/t^{11/2}$ time dependence, respectively, as given by Eq. (D.7). Open circles indicate numerical results calculated by discretization of Eq. (36), while closed circles indicate the numerical results of Shoup and Szabo.²²

where $\text{erfc}(z)$ is the complementary error function, $\text{erfc}(z) = (2/\sqrt{\pi}) \int_z^\infty du \exp(-u^2)$,³³ and c_{pd} is given by

$$c_{pd} = \frac{3\pi(12 - \pi^2)}{8(\pi^2 - 9)} \approx 2.88616. \quad (54)$$

Equation (52) can be readily evaluated using built-in implementations of special functions (*e.g.*, the complementary error function) available in standard scientific software packages.

We compare the result of Eq. (52) with the empirical equation of Shoup and Szabo, reproduced here for completeness,²²

$$\frac{I(t)}{I_\infty} \approx \frac{\pi}{4} + \frac{a}{4} \sqrt{\frac{\pi}{Dt}} + \left(1 - \frac{\pi}{4}\right) \exp \left[-\frac{\sqrt{\pi}/2 - 4\pi^{-3/2}}{1 - \pi/4} \frac{a}{2\sqrt{Dt}} \right]. \quad (55)$$

In Fig. 4, very good agreement is observed among Eq. (52), the Shoup–Szabo empirical equation, and the numerical results. Table II summarizes numerical values of the time dependence of the current for $Dt/a^2 < 1$. A comparison of Tables I and II indicates that Eq. (52) generally

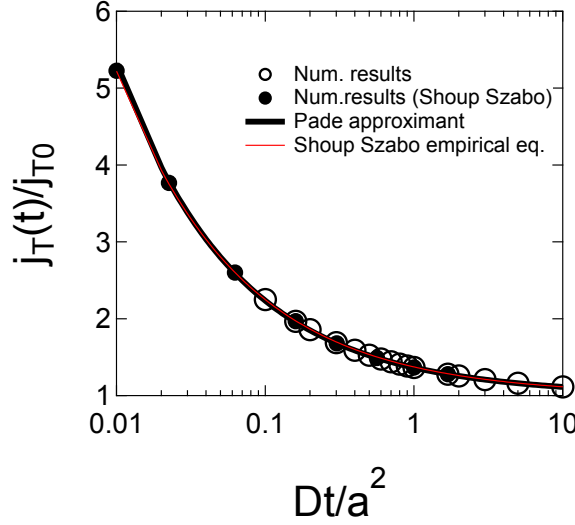


FIG. 4. The black thick solid line shows the result of the Padé approximant [Eq. (52)]. The red thin solid line shows the result of Shoup and Szabo [Eq. (55)].²² Open circles indicate numerical results calculated by discretization of Eq. (36), and closed circles indicate the numerical results of Shoup and Szabo.²²

provides closer agreement with the numerical results than the Shoup–Szabo empirical equation for $Dt/a^2 \gtrsim 0.2$. For shorter times, $Dt/a^2 \lesssim 0.2$, the Shoup–Szabo empirical equation tends to show slightly better agreement. As shown in Appendix E, the typical experimentally accessible time range is $0.01 \lesssim Dt/a^2$, and Eq. (52) can be used to fit experimental data with accuracy comparable to that of the widely used Shoup–Szabo empirical equation [Eq. (55)].

Equation (52) is obtained by extrapolating the asymptotic expansion to shorter times, whereas the Shoup–Szabo empirical equation is constructed by phenomenological interpolation between the short- and long-time limits. Consequently, Eq. (52) is particularly suitable for describing the intermediate- and long-time behavior, while retaining reasonable accuracy in the short-time region.

The results reported by Mahon and Oldham are accurate; however, they are expressed in terms of two separate equations, each applicable to a distinct time regime.²⁴ Analytical expressions based on Padé approximants applied to inverse time-series expansions have also

TABLE II. Time dependence of the normalized current $I(t)/I_0$ for $Dt/a^2 < 1$

Dt/a^2 ^a	High-accuracy numerical ^{b c}	Eq. (52)	Eq. (6) of Mahon–Oldham ²⁴	Eq. (8) of Shoup–Szabo ²²
0.1	2.24999(2.2477)	2.235	2.24761	2.24894
0.2	1.85775(1.8577)	1.84966	1.8574	1.86572
0.3	1.68964(1.6896)	1.68456	1.6892	1.69948
0.5	1.5255(1.5255)	1.52314	1.52528	1.53548
0.7	1.44037(1.4404)	1.43907	1.44052	1.44948
0.9	1.38638(1.3864)	1.38558	1.38675	1.39457

^a Dimensionless time of Shoup–Szabo is $4Dt/a^2$.²²

^b Numerical results calculated using Eq. (36) together with Eq. (38), providing benchmark-level accuracy for the present comparisons.

^c The numerical values in parentheses indicate the results of the numerical simulations reported in Ref. 56.^{44,56,57} The present numerical results are consistent with high-accuracy numerical evaluations reported in the literature (e.g., Ref. 27).

been proposed previously.²⁵ In such approaches, the use of higher-order Padé approximants, such as the [5,4] approximant, naturally leads to more involved expressions containing a larger number of terms.²⁵ Alternatively, a simpler approximate expression has been proposed using empirical considerations.⁵⁸

VIII. SHORT-TIME REGIME

Compared with the long-time asymptotic behavior, the short-time relaxation of the current is more difficult to analyze because an instantaneous change in interfacial ion concentration leads to discontinuities in the current density near the edge of the disk electrode.²⁷ These features originate from the mixed boundary-value nature of the problem, in which Dirichlet boundary conditions apply on the electrode surface, whereas Neumann boundary conditions apply on the surrounding insulating plane. Such a configuration is characteristic of mixed boundary-value (Zaremba-type) problems.^{27,59,60}

A. Mathematical analysis.

The integral equation [Eq. (36)] with the kernel given by Eq. (38) reproduces results obtained from rigorous formulations based on radial spheroidal wave functions of the third kind.^{25–27} As shown in Appendix D, the numerical coefficients of the power-law expansion of the current density in Eq. (D.8), up to $(a/\sqrt{Dt})^{13}$, are in exact agreement with those reported in Table 1 of Ref. 27. Since Eq. (36) with Eq. (38) is derived from Eq. (35), we employ Eq. (35) to examine the short-time behavior. This regime can be analyzed by considering the limit $s \rightarrow \infty$, in which Eq. (35) is approximated as

$$\frac{2}{\pi} \int_0^a dy \hat{h}(y, s) \int_0^\infty dk \cos(ky) \frac{k}{\sqrt{s/D}} J_0(kr) \approx \frac{1}{s} \quad \text{for } 0 \leq r < a. \quad (56)$$

Using 10.22.59 of Ref. 33 (1.4.13 of Ref. 32),

$$\int_0^\infty dk \sin(ky) J_0(kr) = \begin{cases} 1/\sqrt{y^2 - r^2}, & y > r, \\ 0, & 0 \leq y < r, \end{cases} \quad (57)$$

we obtain the corresponding cosine transform by differentiating with respect to y (Abel-regularized),

$$\int_0^\infty dk k \cos(ky) J_0(kr) = \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{y^2 - r^2}} \right) \quad (y > r), \quad (58)$$

and the left-hand side vanishes for $0 \leq y < r$. Substitution of Eq. (58) into Eq. (56) yields

$$\lim_{\epsilon \rightarrow +0} \frac{2}{\pi} \int_{r+\epsilon}^a dy \hat{h}(y, s) \frac{\partial}{\partial y} \frac{1}{\sqrt{y^2 - r^2}} \approx \frac{1}{\sqrt{sD}}. \quad (59)$$

Performing partial integration gives

$$\lim_{\epsilon \rightarrow +0} \frac{2}{\pi} \left[\hat{h}(y, s) \frac{1}{\sqrt{y^2 - r^2}} \Big|_{y=r+\epsilon}^{y=a} - \int_r^a dy \frac{\partial \hat{h}(y, s)}{\partial y} \frac{1}{\sqrt{y^2 - r^2}} \right] \approx \frac{1}{\sqrt{sD}}. \quad (60)$$

Apart from the singular contribution associated with the inner endpoint $y = r$, Eq. (60) reduces to the integral term involving $\partial \hat{h}/\partial y$. A regular short-time solution consistent with this integral equation is given by (see Appendix F)

$$\hat{h}(y, s) = \frac{\sqrt{a^2 - y^2}}{\sqrt{sD}}, \quad (61)$$

which satisfies the integral term in Eq. (60) and yields the required right-hand side $1/\sqrt{sD}$.

Substitution of Eq. (61) into $\hat{I}(s) = 4DnFC_0 \int_0^a dr \hat{h}(r, s)$ [Eq. (23)] yields $\hat{I}(s) = DnFC_0\pi a^2/\sqrt{sD}$. Upon inverse Laplace transformation ($1/\sqrt{s} \leftrightarrow 1/\sqrt{\pi t}$), this result leads to Cottrell's equation,^{5,18–23,61,62}

$$I(t) = nF\pi a^2 C_0 \sqrt{\frac{D}{\pi t}}.$$

This analysis demonstrates that the integral equation admits at least one admissible short-time solution that is integrable over $0 \leq r < a$ and yields a finite total current. Since the experimentally observable quantity is the spatially integrated current, any local modification of $\hat{h}(r, s)$ that does not alter this integral does not affect the leading-order short-time behavior of $\hat{I}(s)$.

B. Physical interpretation.

In one-dimensional diffusion, Cottrell's equation is derived under the implicit assumption of a uniform current density over the electrode surface. In contrast, the present solution exhibits an explicit radial dependence of $\hat{h}(r, s)$ and involves a kernel singularity at $y = r$ in $\hat{K}(r, y, s)$, which is characteristic of mixed boundary-value problems. The normal component of the flux density exhibits a radial dependence proportional to $\sqrt{a^2 - r^2}$ and vanishes at the disk edge, reflecting the Neumann boundary condition on the surrounding insulating plane. At the initial instant, the discontinuous change in interfacial ion concentration implies that radial transport along the electrode surface cannot be excluded, particularly near the disk periphery. Nevertheless, the present analysis demonstrates that the total current follows Cottrell's equation, consistent with previous analytical^{5,18–23,62} and numerical results.⁵⁶

It is worth noting that, although the formal short-time limit yields Cottrell's equation, this regime corresponds to an idealized instantaneous change in interfacial concentration induced by a potential step and is often experimentally obscured by finite instrumental response times, double-layer charging, and other non-diffusive effects.^{5,31} In contrast, the long-time asymptotic expansion describes the relaxation toward the finite steady-state current $I_\infty \propto a$. The excess current follows a Cottrell-type decay, $I(t) - I_\infty \propto a^2 \sqrt{D/t}$, with a distinct amplitude, reflecting

the three-dimensional expansion of the diffusion field required to maintain steady-state transport to a finite disk. This regime typically governs practical chronoamperometric measurements at disk and microdisk electrodes and underlies widely used analytical approximations.⁸

IX. RELATION TO DIFFUSIONAL EC' MECHANISM

Owing to the initial condition $\Delta C(r, z, 0) = 0$, the diffusion equation in the Laplace domain, Eq. (A.1), can be formally interpreted as a diffusion equation with a quasi-first-order reaction,

$$D\nabla^2\Delta C = K\Delta C(r, z), \quad (62)$$

by identifying the Laplace variable s with K in the steady-state limit.^{25,63}

A steady-state quantity can be obtained from its Laplace-domain counterpart by multiplication by the Laplace variable s . Accordingly, the steady-state current in the presence of K is given by $I_K = s\hat{I}(s)\big|_{s=K}$. Using Eqs. (50) and (51), we obtain

$$\frac{I_K}{I_\infty} \approx 1 + \frac{2a}{\pi} \sqrt{\frac{K}{D}} + \left(\frac{4}{\pi^2} - \frac{1}{3}\right) \frac{a^2 K}{D} - \left(\frac{1}{9} - \frac{1}{\pi^2}\right) \frac{8a^3 K^{3/2}}{\pi D^{3/2}} \quad (63)$$

$$\approx \frac{3\pi(12 - \pi^2) + 2\pi^2 a \sqrt{K/D} + \pi(\pi^2 - 8) a^2 K/D}{3\pi(12 - \pi^2) + 8(\pi^2 - 9) a \sqrt{K/D}}. \quad (64)$$

Equation (63) has been derived previously.²⁵ Results based on higher-order Padé approximants have also been reported and shown to provide accurate descriptions of the current.^{25,64–66}

Equation (62) is mathematically identical to the reaction–diffusion equation commonly used to describe a diffusional EC' (electrochemical–catalytic or regenerative) mechanism at a disk electrode. In this framework, the electroactive species undergoes a homogeneous reaction characterized by a quasi-first-order rate constant K .^{25,63} In the classical EC' framework, this reaction term gives rise to a steady-state current determined by the competition between diffusion and reaction rates.^{67–69}

Within the present formulation, identifying the Laplace variable s with the reaction rate constant K in the steady-state limit establishes a formal correspondence between the time-dependent diffusion problem and the steady-state EC' description.^{25,63,70} Accordingly, the

steady-state current I_K obtained from Eq. (64) can be interpreted as providing analytical expressions that are formally analogous to the EC' steady-state current at a disk electrode. The ratio I_K/I_∞ therefore provides a measure analogous to the catalytic enhancement characteristic of an EC' mechanism, relative to the purely diffusion-controlled current.

In the limiting case $K \rightarrow 0$, the reaction becomes negligible and $I_K/I_\infty \rightarrow 1$, recovering the diffusion-controlled behavior. Conversely, in the limit $K \rightarrow \infty$, rapid chemical regeneration leads to an enhanced steady-state current, consistent with the catalytic EC' regime discussed in classical analyses.^{25,65} This behavior confirms that the present formulation reproduces the expected limiting behavior associated with diffusional EC' mechanisms, while providing a continuous analytical description over intermediate values of K .

X. CONCLUSION

We have investigated the relaxation of the diffusion-limited current at a disk electrode following a change in interfacial ion concentration induced by a potential step over a circular reactive region embedded in an otherwise insulating plane. Starting from the diffusion equation with mixed Dirichlet–Neumann boundary conditions, the problem was formulated in the Laplace domain as a modified Helmholtz equation. The resulting dual integral equations were reduced, via the Cooke–Sneddon method, to a single Fredholm integral equation for an auxiliary function $\hat{h}(r, s)$. This function directly determines the total Faradaic current, thereby providing a transparent link between the mixed boundary-value formulation and the observable current response.

In the steady-state limit, Saito's equation was recovered. For the transient response, two complementary approaches were developed. Numerical discretization of the Fredholm equation, combined with numerical Laplace inversion, yields accurate results for $Dt/a^2 \gtrsim 0.1$. In parallel, a systematic long-time asymptotic expansion generates inverse-power-series corrections to the steady-state current, reproducing known results and providing higher-order terms within a unified framework. Because convergence is more favorable in the Laplace domain, a Padé resummation was introduced, leading to a simple closed-form approximation with excellent

agreement with numerical results.

The short-time analysis recovers Cottrell’s equation while highlighting the role of edge effects associated with the mixed boundary conditions. Furthermore, identifying the Laplace variable with a quasi-first-order reaction rate constant establishes a direct connection between the present transient formulation and diffusional EC’ mechanisms.

Although the formal short-time limit yields area-proportional Cottrell’s equation, this regime corresponds to an idealized instantaneous step and is often obscured experimentally. In contrast, the long-time asymptotic correction follows Cottrell-type decay with a distinct amplitude associated with relaxation toward the finite steady-state current of a disk electrode. This transient reflects the three-dimensional evolution of the diffusion field and typically governs practical chronoamperometric measurements.

Overall, the integral-equation framework developed here provides a unified and rigorous description of transient diffusion-limited currents at disk electrodes, bridging steady-state and time-dependent behavior and offering reliable tools for quantitative electrochemical analysis.

APPENDIX A. DERIVATION OF EQ. (4)

In the Laplace domain, we have

$$s\Delta\hat{C}(r, z, s) = D\nabla^2\Delta\hat{C}, \quad (\text{A.1})$$

where $\Delta C(r, z, 0) = 0$ is used. Equation (A.1) yields

$$\nabla^2\Delta\hat{C}(r, z, s) - \frac{s}{D}\Delta\hat{C}(r, z, s) = \Delta_2\Delta\hat{C} + \frac{\partial^2}{\partial z^2}\Delta\hat{C} - \frac{s}{D}\Delta\hat{C} = 0. \quad (\text{A.2})$$

In the Laplace domain, the diffusion equation with zero initial conditions is the modified Helmholtz equation, $\Delta_2\Delta\hat{C} + \frac{\partial^2}{\partial z^2}\Delta\hat{C} = \kappa^2\Delta\hat{C}$, where $\kappa^2 = s/D$.

We introduce $\Psi(r)$ satisfying $\Delta_2\Psi(r) = -\kappa^2\Psi(r)$, which gives $\Psi(r) = J_0(\kappa r)$ by noting that the Bessel equation can be written as

$$\frac{1}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r}J_n(\kappa r) + \kappa^2J_n(\kappa r) - \frac{n^2}{r^2}J_n(\kappa r) = 0, \quad (\text{A.3})$$

where $J_n(z)$ indicates the Bessel functions of the first kind, and the solution should be finite at $r = 0$.³³

First, we consider the solution without taking into account the boundary conditions for the circular domain of radius a at $z = 0$. Assuming separation of variables, $\Delta\hat{C}(r, z, s) = \Psi(r)\Phi(z, s)$, we substitute this form into Eq. (A.2) and divide by $\Psi(r)\Phi(z, s)$, which yields

$$\frac{\Delta_2\Psi}{\Psi} = -\frac{1}{\Phi}\frac{\partial^2}{\partial z^2}\Phi + \frac{s}{D} = -\kappa^2, \quad (\text{A.4})$$

and $\Phi(z, s) = \exp\left(-|z|\sqrt{\kappa^2 + s/D}\right)$. Therefore, the solution of Eq. (A.1) in the absence of boundary conditions can be written as

$$\Delta\hat{C}(r, z, s) = J_0(\kappa r) \exp\left(-|z|\sqrt{\kappa^2 + s/D}\right). \quad (\text{A.5})$$

It is readily verified by direct substitution that Eq. (A.5) satisfies Eq. (A.1). In Eq. (4), $A(k, s)$ is introduced to enforce the boundary conditions.

APPENDIX B: DERIVATION OF EQ. (36)

Using the integral representation of the Bessel function $J_0(kr)$ ³³

$$J_0(kr) = \frac{2}{\pi} \int_0^r du \frac{\cos(ku)}{\sqrt{r^2 - u^2}}, \quad (\text{B.1})$$

we rewrite Eq. (35) as

$$\int_0^r du \frac{G(u)}{\sqrt{r^2 - u^2}} = \frac{1}{s}, \quad (\text{B.2})$$

where $G(u)$ is defined as

$$G(u) \equiv \left(\frac{2}{\pi}\right)^2 \int_0^a dy \hat{h}(y, s) \int_0^\infty dk \frac{k}{\sqrt{k^2 + s/D}} \cos(ku) \cos(ky). \quad (\text{B.3})$$

The Abel-type integral equation in Eq. (B.2) can be solved by a standard method.^{32,35,71} We multiply both sides of Eq. (B.2) by $\int_0^x dr \frac{2r}{\sqrt{x^2 - r^2}}$. From the left-hand side, we obtain

$$\int_0^x dr \frac{2r}{\sqrt{x^2 - r^2}} \int_0^r du \frac{G(u)}{\sqrt{r^2 - u^2}} = \int_0^x du G(u) \int_u^x dr \frac{2r}{\sqrt{x^2 - r^2} \sqrt{r^2 - u^2}} \quad (\text{B.4})$$

$$= \int_0^x du G(u) \int_0^1 dw \frac{1}{\sqrt{1-w} \sqrt{w}} \quad (\text{B.5})$$

$$= B(1/2, 1/2) \int_0^x du G(u) = \pi \int_0^x du G(u), \quad (\text{B.6})$$

where we introduced $w = (r^2 - u^2)/(x^2 - u^2)$; $B(p, q)$ is the Beta function, and we have used $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ with $\Gamma(1/2) = \sqrt{\pi}$.³³ By differentiating with respect to x , we obtain^{32,35}

$$G(x) = \frac{2}{\pi} \frac{d}{dx} \left[\int_0^x dr \frac{r/s}{\sqrt{x^2 - r^2}} \right] = \frac{2}{\pi s}. \quad (\text{B.7})$$

By combining Eqs. (B.3) and (B.7), we obtain³²

$$\frac{2}{\pi} \int_0^a dy \hat{h}(y, s) \int_0^\infty dk \frac{k}{\sqrt{k^2 + s/D}} \cos(ky) \cos(kr) = \frac{1}{s} \text{ for } 0 \leq r < a. \quad (\text{B.8})$$

$\hat{h}(y, s)$ should be a solution of Eq. (B.8) in order to satisfy the boundary condition in Eq. (18), originating from Eq. (13).

Below, we rewrite Eq. (B.8) into a form suitable for obtaining a numerical solution. Using (1.17.12.1 of Ref. 72 or 2.1.18 of Ref. 32),

$$\delta(r - y) = \frac{2}{\pi} \int_0^\infty dk \cos(kr) \cos(ky), \quad (\text{B.9})$$

we obtain

$$\begin{aligned} \int_0^a dy h(y) \frac{2}{\pi} \int_0^\infty dk \cos(ky) \cos(kr) &= \int_0^a dy h(y) \delta(r - y) \\ &= \begin{cases} h(r), & 0 \leq r < a, \\ 0, & r > a. \end{cases} \end{aligned} \quad (\text{B.10})$$

By combining Eqs. (B.8) and (B.10), we obtain Eq. (36).

APPENDIX C: DIMENSIONLESS FORMULATION

For generality, we reformulate the problem using the dimensionless variables $x = r/a$ and $\xi = ka$. Equation (4) can be expressed using $z_n = z/a$ as,

$$\Delta \hat{C}(r, z, s) = \int_0^\infty d\xi \bar{A}(\xi, s) J_0(\xi x) \exp\left(-z_n \sqrt{\xi^2 + sa^2/D}\right). \quad (\text{C.1})$$

Similarly, the equations corresponding to Eqs. (17)–(19) are expressed as,

$$\bar{g}(\xi, s) = \sqrt{\xi^2 + sa^2/D} \bar{A}(\xi, s)/C_0, \quad (\text{C.2})$$

$$\int_0^\infty d\xi L(\xi, s) \bar{g}(\xi, s) J_\nu(\xi x) = x^\nu \text{ for } 0 \leq x < 1, \quad (\text{C.3})$$

$$\int_0^\infty d\xi \bar{g}(\xi, s) J_\nu(\xi x) = 0 \text{ for } 1 < x < \infty, \quad (\text{C.4})$$

where the Laplace transform of a function expressed in dimensionless variables (*e.g.*, f) is denoted by $\bar{f}$. In our case, $L(\xi, s) \equiv s/\sqrt{\xi^2 + sa^2/D}$ and $\nu = 0$.

According to Cooke (1956) (4.3.24–4.3.27, p. 213 of Ref. 32),³⁴ the solution of Eqs. (C.3) and (C.4) is given by

$$\bar{g}(\xi, s) = \frac{2^\beta \Gamma(\nu + 1)}{\Gamma(\nu - \beta + 1)} \xi^{1+\beta} \int_0^1 dt \bar{f}(t, s) t^{\alpha+1} J_{\nu-\beta}(\xi t). \quad (\text{C.5})$$

where $\Gamma(z)$ is the Gamma function;³³ $\bar{f}(x, s)$ satisfies

$$\bar{f}(x, s) + x^{-\alpha} \int_0^1 dt t^{\alpha+1} \bar{f}(t, s) \left[\int_0^\infty d\xi (a\xi^{2\beta} L(\xi, s) - 1) \xi J_{\nu-\beta}(\xi t) J_{\nu-\beta}(\xi x) \right] = ax^{\nu-\alpha-\beta}, \quad (\text{C.6})$$

where a , α , and β are arbitrary parameters, provided that $0 < \beta < 1$ and $-1 < \nu - \beta$ for real β and ν .

We choose $\beta = 1/2$, so that $J_{-1/2}(\xi t) = \sqrt{2/(\pi \xi t)} \cos(\xi t)$. The other parameters are chosen as $a = 1/s$ and $\alpha = -1/2$.

Accordingly, we introduce $\bar{h}(x, s)$ satisfying

$$\bar{g}(\xi, s) = \frac{2}{\pi} \xi \int_0^1 dx \bar{h}(x, s) \cos(\xi x), \quad (\text{C.7})$$

and

$$\bar{h}(x, s) + \frac{2}{\pi} \int_0^1 dy \bar{h}(y, s) \int_0^\infty d\xi \left[\frac{\xi}{\sqrt{\xi^2 + sa^2/D}} - 1 \right] \cos(\xi y) \cos(\xi x) = \frac{1}{s} \text{ for } 0 \leq x < 1. \quad (\text{C.8})$$

Using the dimensionless formulation, Eqs. (21) and (23) become

$$\hat{I}(s) = 2\pi D a n F C_0 \int_0^\infty d\xi \bar{g}(\xi, s) \frac{1}{\xi} J_1(\xi), \quad (\text{C.9})$$

$$= 4D n F C_0 \int_0^1 dx \bar{h}(x, s). \quad (\text{C.10})$$

APPENDIX D: FAST EVALUATION OF THE ITERATED FUNCTIONS AND THE TOTAL CURRENT

The recursive relation for the iterated functions,

$$h^{(\ell)}(r) = \sum_{n=1}^{\ell} \int_0^a dy \bar{K}^{(n)}(r, y) h^{(\ell-n)}(y), \quad \ell \geq 1, \quad (\text{D.1})$$

with $h^{(0)}(r) = 1$ and $\bar{K}^{(n)} = K^{(n)}/a^n$, is identical to Eq. (44) in the main text. A direct symbolic implementation of this recursion is computationally expensive, since each iteration involves repeated symbolic integrations. In this Appendix, we present an efficient computational scheme that exploits the polynomial structure of the iterated functions and avoids symbolic integration altogether.

Since $K^{(n)}(r, y)$, given by Eq. (40), is a polynomial in r with maximum degree $n - 1$, it follows that each iterated function $h^{(\ell)}(r)$ is also a polynomial in r with maximum degree ℓ , given that $h^{(0)}(r) = 1$. Accordingly, we may write

$$h^{(\ell)}(r) = \sum_{m=0}^{\ell} c_m r^m.$$

Furthermore, the kernel $\bar{K}^{(n)}(r, y)$ is a linear combination of the polynomial factors $(r + y)^{n-1}$, $(r - y)^{n-1}$, and $|r - y|^{n-1}$ [cf. Eq. (40)]. By expanding these factors binomially, the integrand in Eq. (D.1) is reduced to a finite sum of monomials in y .

As a consequence, all integrals appearing in the recursion can be expressed in terms of the elementary moments

$$M_a(m) = \int_0^a y^m dy = \frac{a^{m+1}}{m+1}, \quad (\text{D.2})$$

$$M_r(m) = \int_0^r y^m dy = \frac{r^{m+1}}{m+1}, \quad (\text{D.3})$$

$$M_{r \rightarrow a}(m) = \int_r^a y^m dy = \frac{a^{m+1} - r^{m+1}}{m+1}. \quad (\text{D.4})$$

The absolute-value contribution is treated exactly by splitting the integration domain at $y = r$,

$$\int_0^a |r - y|^{n-1} y^m dy = \int_0^r (r - y)^{n-1} y^m dy + \int_r^a (y - r)^{n-1} y^m dy,$$

after which each term is again reduced to the precomputed moments M_r and $M_{r \rightarrow a}$ following binomial expansion.

In practice, the evaluation of the right-hand side of Eq. (D.1) for a fixed n proceeds as follows: (i) expand $h^{(\ell-n)}(y)$ into monomials; (ii) expand the kernel factors $(r \pm y)^{n-1}$ and $(y - r)^{n-1}$ binomially; and (iii) replace all resulting integrals of y^m over the relevant interval by $M_a(m)$, $M_r(m)$, or $M_{r \rightarrow a}(m)$. This procedure yields an explicit polynomial expression for $h^{(\ell)}(r)$ without invoking symbolic integration. Iterating this procedure efficiently implements the recursion (D.1) and enables the computation of high-order iterated functions.

Once $h^{(\ell)}(r)$ has been obtained in polynomial form, the coefficient entering the asymptotic expansion of the normalized current is evaluated as

$$\frac{I^{(\ell)}}{I_\infty} = \frac{1}{a} \int_0^a h^{(\ell)}(r) dr,$$

where $I_\infty = 4DanFC_0$. This integral is again computed using the monomial rule $\int_0^a r^m dr = a^{m+1}/(m+1)$. The truncated long-time expansion of the normalized current is then constructed using,

$$\frac{I(t)}{I_\infty} \approx 1 + \sum_{\ell=0}^{\ell_{\max}} \frac{I^{(1+2\ell)}}{I_\infty} \frac{(Dt/a^2)^{-1/2-\ell}}{\Gamma(\frac{1}{2}-\ell)},$$

while the corresponding expression in the Laplace domain is

$$\frac{\hat{I}(s)}{I_\infty} \approx \frac{1}{s} \left(1 + \sum_{\ell=0}^{\ell_{\max}} \frac{I^{(1+2\ell)}}{I_\infty} (sa^2/D)^{1/2+\ell} \right). \quad (\text{D.5})$$

As an illustration of the efficiency of this approach, we report the systematic asymptotic expansion up to a certain order in the Laplace domain,

$$\begin{aligned} \frac{\hat{I}(s)}{I_\infty} \approx & \frac{1}{s} + \frac{2a}{\pi(Ds)^{1/2}} - \left(\frac{1}{9} - \frac{1}{\pi^2}\right) \frac{8a^3 s^{1/2}}{\pi D^{3/2}} + \left(\frac{142}{675} - \frac{16}{3\pi^2} + \frac{32}{\pi^4}\right) \frac{a^5 s^{3/2}}{\pi D^{5/2}} + \\ & \left(-\frac{1136}{33075} + \frac{3872}{2025\pi^2} - \frac{256}{9\pi^4} + \frac{128}{\pi^6}\right) \frac{a^7 s^{5/2}}{\pi D^{7/2}} + \left(\frac{19046}{4465125} - \frac{86336}{178605\pi^2} + \frac{608}{45\pi^4} - \frac{1280}{9\pi^6} + \frac{512}{\pi^8}\right) \frac{a^9 s^{7/2}}{\pi D^{9/2}} + \\ & \left(-\frac{658136}{1620840375} + \frac{1280408}{13395375\pi^2} - \frac{161408}{35721\pi^4} + \frac{56576}{675\pi^6} - \frac{2048}{3\pi^8} + \frac{2048}{\pi^{10}}\right) \frac{a^{11} s^{9/2}}{\pi D^{11/2}} + \\ & \left(\frac{57122836}{1917454163625} - \frac{400816}{25727625\pi^2} + \frac{34023424}{28704375\pi^4} - \frac{4417024}{127575\pi^6} \right. \\ & \left. + \frac{971264}{2025\pi^8} - \frac{28672}{9\pi^{10}} + \frac{8192}{\pi^{12}}\right) \frac{a^{13} s^{11/2}}{\pi D^{13/2}}. \quad (\text{D.6}) \end{aligned}$$

Applying the inverse Laplace transformation via the Tauberian theorem yields the corresponding expression in the time domain,

$$\begin{aligned} \frac{I(t)}{I_\infty} \approx & 1 + \frac{2a}{\pi^{3/2}(Dt)^{1/2}} + \left(\frac{1}{9} - \frac{1}{\pi^2}\right) \frac{4a^3}{(\pi Dt)^{3/2}} + \left(\frac{71}{450} - \frac{4}{\pi^2} + \frac{24}{\pi^4}\right) \frac{a^5}{\pi^{3/2}(Dt)^{5/2}} + \\ & \left(\frac{71}{2205} - \frac{242}{135\pi^2} + \frac{80}{3\pi^4} - \frac{120}{\pi^6}\right) \frac{2a^7}{\pi^{3/2}(Dt)^{7/2}} + \left(\frac{9523}{340200} - \frac{5396}{1701\pi^2} + \frac{266}{3\pi^4} - \frac{2800}{3\pi^6} + \frac{3360}{\pi^8}\right) \frac{a^9}{\pi^{3/2}(Dt)^{9/2}} + \\ & \left(\frac{82267}{6860700} - \frac{160051}{56700\pi^2} + \frac{25220}{189\pi^4} - \frac{12376}{5\pi^6} + \frac{20160}{\pi^8} - \frac{60480}{\pi^{10}}\right) \frac{a^{11}}{\pi^{3/2}(Dt)^{11/2}} + \\ & \left(\frac{14280709}{2951348400} - \frac{25051}{9900\pi^2} + \frac{5847776}{30375\pi^4} - \frac{759176}{135\pi^6} + \frac{1168552}{15\pi^8} - \frac{517440}{\pi^{10}} + \frac{1330560}{\pi^{12}}\right) \frac{a^{13}}{\pi^{3/2}(Dt)^{13/2}}. \quad (\text{D.7}) \end{aligned}$$

In order to compare Eq. (D.7) with previous results, we express Eq. (D.7) by truncating the irrational numbers originating from π as

$$\begin{aligned} \frac{I(t)}{I_\infty} \approx & 1 + 0.359174 \sqrt{\frac{a^2}{Dt}} + 0.00703258 \left(\frac{a^2}{Dt}\right)^{3/2} - 0.000201745 \left(\frac{a^2}{Dt}\right)^{5/2} - 0.000175244 \left(\frac{a^2}{Dt}\right)^{7/2} + \\ & 0.0000217006 \left(\frac{a^2}{Dt}\right)^{9/2} + 0.0000195802 \left(\frac{a^2}{Dt}\right)^{11/2} - 5.50927 \times 10^{-6} \frac{a^{13}}{(Dt)^{13/2}}. \quad (\text{D.8}) \end{aligned}$$

The numerical values are identical to those reported in Table 1 of Ref. 27. In Ref. 27, these values are obtained by evaluating the exact expression for the current expanded in terms of radial spheroidal wave functions of the third kind.^{25–27} The agreement between our results and those

reported in Ref. 27 indicates the correctness of our calculations. The terms from $1/t^{5/2}$ up to the $1/t^{9/2}$ time dependence have also been reported numerically.^{23,24} Our results are consistent, up to the first three digits of the coefficient of $1/t^{5/2}$, with Eq. (6) of Mahon and Oldham²⁴, after multiplying the coefficient by $4/\pi$ to account for the difference in the normalization of the current; according to the normalization reported in Ref. 24, the numerical coefficients are $4/\pi$, $8/\pi^{5/2}$, 0.00895416, 0.00025687, 0.000223127, 0.0000276301, and 0.0000249303.

APPENDIX E: EXPERIMENTAL VERIFICATION OF THE TRANSIENT CURRENT PREDICTION OF EQ. (52)

Here, we present an experimental verification of the transient current predictions using reported potential step chronoamperometry data shown in Fig. 6(a) of Ref. 73. We analyze dpotential step chronoamperometry data of a typical reversible redox couple ($[\text{Fe}(\text{CN})_6]^{4-}/[\text{Fe}(\text{CN})_6]^{3-}$) (1 mol m^{-3}) in aqueous KCl solution (1 mol dm^{-3}) at a Pt disk electrode (radius 0.101 mm) at 25 °C. The other experimental conditions are given in Ref. 73.

The diffusion coefficients and the transient currents analyzed using Eq. (52) are compared with those estimated using the Shoup–Szabo equation [Eq. (55)]. The experimentally accessible time range corresponds to $0.01 \lesssim Dt/a^2 \lesssim 0.2$, in which Eq. (52) is expected to provide sufficient accuracy, as demonstrated in Fig. 4. Note that Eq. (52) is constructed using a Padé approximant that refines the long-time asymptotic decay, whereas the Shoup–Szabo equation interpolates between Cottrell’s equation (valid in the short-time regime) and the long-time asymptotic decay. In the present time range, the short-time regime described by Cottrell’s equation is not dominant (it typically appears for $Dt/a^2 \lesssim 0.01$), making the Padé-based expression conceptually more suitable.

In both Eqs. (52) and (55), we introduce a single fitting parameter $C_F = D/a^2$ and express $I_\infty = 4DanFC_0$ as $I_\infty = 4C_F a^3 nFC_0$.

Both models were fitted to the data in the time window $0.20 \leq t \leq 4.00 \text{ s}$, excluding the unresolved early-time transient region.

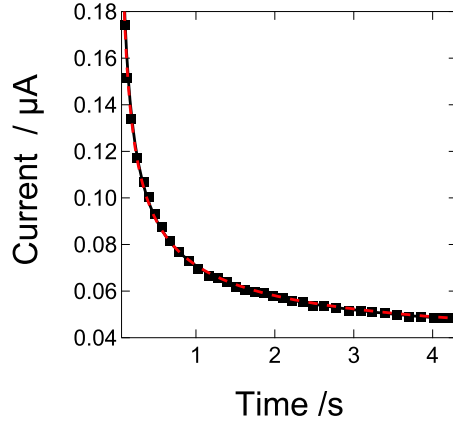


FIG. 5. The dots represent the absolute value of the cathodic current transient for potential step chronoamperometry of $[\text{Fe}(\text{CN})_6]^{3-}$ (1 mol m^{-3}) in aqueous KCl solution (1 mol dm^{-3}) at a Pt disk electrode (radius 0.101 mm) at 25°C , reproduced from Ref. 73 (Fig. 6(a)). The black thick solid line shows the fit to the Padé approximant [Eq. (52)], and the red thick dashed line shows the fit to the Shoup–Szabo equation [Eq. (55)]²². Reproduced from Ref. 73 with permission from Elsevier.

The diffusion coefficients obtained from the two models are:

$$D = (7.35 \pm 0.02) \times 10^{-10} \text{ m}^2 \text{ s}^{-1}, \quad (\text{E.1})$$

$$D = (7.33 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1}, \quad (\text{E.2})$$

using Eq. (52) and the Shoup–Szabo equation, respectively, where the uncertainty was obtained from the standard deviation of the residuals within the fitting window.

Accordingly, both models reproduce the experimental data with comparable accuracy in the measured time range, as shown in Fig. 5. The remaining deviations are attributed to digitization of the original figure and the unresolved early-time transient.

For comparison, the diffusion coefficient reported in the original experimental work is $7.74 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$, averaged over four measurements⁷³. This value is in good agreement with those obtained here, confirming that both models provide reliable estimates of the diffusion coefficient within the experimentally accessible time range.

APPENDIX F: VERIFICATION OF EQ. (61)

To verify that Eq. (61) satisfies the regular (integral) term in Eq. (60), we note that

$$\frac{\partial \hat{h}(y, s)}{\partial y} = -\frac{1}{\sqrt{sD}} \frac{y}{\sqrt{a^2 - y^2}}. \quad (\text{F.1})$$

Therefore,

$$-\int_r^a dy \frac{\partial \hat{h}(y, s)}{\partial y} \frac{1}{\sqrt{y^2 - r^2}} = \frac{1}{\sqrt{sD}} \int_r^a dy \frac{y}{\sqrt{a^2 - y^2}} \frac{1}{\sqrt{y^2 - r^2}}. \quad (\text{F.2})$$

Using the integral formula

$$\int_r^a dy \frac{y}{\sqrt{a^2 - y^2}} \frac{1}{\sqrt{y^2 - r^2}} = \frac{\pi}{2}, \quad (\text{F.3})$$

obtained by the change of variables $t = y^2$ and the Beta-function identity (Eq. (3.196.3) of Ref. 38)³³

$$\int_\alpha^\beta dt (\beta - t)^{\nu-1} (t - \alpha)^{\mu-1} = (\beta - \alpha)^{\mu+\nu-1} B(\mu, \nu), \quad (\text{F.4})$$

with $\mu = \nu = 1/2$ and $B(1/2, 1/2) = \Gamma(1/2)^2 = \pi$, Eq. (F.2) becomes

$$-\int_r^a dy \frac{\partial \hat{h}(y, s)}{\partial y} \frac{1}{\sqrt{y^2 - r^2}} = \frac{\pi}{2} \frac{1}{\sqrt{sD}}.$$

Substituting this result into Eq. (60) yields $1/\sqrt{sD}$ for the regular contribution, which is consistent with Eq. (61).

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